

Nexus Broadcast — UI Design Specification for Cursor

This document is a complete, authoritative reference for the Nexus Broadcast app UI. Use it as the single source of truth when implementing, extending, or modifying any screen. Every design decision is intentional — do not deviate without a strong reason.

1. Design Philosophy

Theme: Obsidian Studio — Professional Broadcast Hardware UI

The design is inspired by high-end broadcast control surfaces: Blackmagic ATEM switchers, Ross Video production systems, and DaVinci Resolve. The goal is to feel like **a physical piece of broadcast equipment rendered in software** — not a generic dark SaaS dashboard.

Every screen must feel like a **control room panel**: dense, purposeful, always showing operational state. Empty space is not “clean” — it is “waiting for signal.” Fill it with instrumentation.

Four Core Principles:

1. **Information density without clutter** — every pixel earns its place; no decorative whitespace
2. **Hierarchy through luminosity** — brighter = more important; dim = secondary/inactive
3. **Precision over decoration** — functional beauty; no gradients unless they carry meaning
4. **Status at a glance** — critical states (LIVE, errors, health) are always visible

2. Color System

All colors are defined as exact hex or rgba values. Do **not** substitute with Tailwind color names — always use inline styles or CSS variables for these.

Surface Hierarchy

Layer	Color	Usage
App background	#0E0F14	Root background, main content area
Sidebar / top-level panels	#111318	Sidebar, panel backgrounds
Elevated panels / cards	#16181F	Secondary panels, settings tabs
Input / control backgrounds	#0E0F14	Text inputs, selects, buttons
Active/selected state	#1A1D2B	Highlighted nav item, selected row
Canvas / workspace	#0A0B0F	Program output canvas, scene editor workspace

Borders

All borders are **whisper-thin** — they define structure without adding visual weight.

Primary border:

rgba(255,255,255,0.07)

– panel edges, sidebar border

Secondary border:

rgba(255,255,255,0.08)

– card/panel borders

Subtle divider:

rgba(255,255,255,0.05)

– inner section dividers

Input border:

rgba(255,255,255,0.12)

– form inputs (slightly more v:

Accent Colors (Semantic — Never Swap These)

Color	Hex	Semantic Meaning	Use For
Nexus Blue	#3B82F6	Primary interaction / brand	Active states, CTAs, selected items, primary buttons, active nav border
Live Red	#EF4444	Danger / live state	LIVE badges, End Stream button, critical alerts, dropped frames
Health Green	#22C55E	Good / connected	Signal OK, CPU healthy, connected status, success states
Warning Amber	#F59E0B	Caution / degraded	Low bitrate, high CPU, donation highlights
Cyan	#06B6D4	Secondary data / timecode	Timecodes, chart secondary lines, uptime counters
Muted White	rgba(255, 255, 255, 0.5)	Secondary labels	Inactive nav items, secondary text
Dim White	rgba(255, 255, 255, 0.3)	Tertiary labels	Metadata, timestamps, placeholder text
Ghost White	rgba(255, 255, 255, 0.1)	Decorative	Subtle borders, disabled states

Rule: Third-party platform colors (Twitch #9146FF, YouTube #FF0000, Facebook #1877F2) are allowed **only** for platform-selector buttons and chat username colors. They must never dominate a panel — always subordinate to Nexus Blue.

3. Typography

Three fonts are loaded via Google Fonts. All three serve distinct roles — never substitute one for

another.

```
<link href="https://fonts.googleapis.com/css2?family=Space+Grotesk:wght@500;700;900&display=block">
```

Font Roles

Font	Role	Where Used
Space Grotesk	Display / Brand	Page titles, section headers, logo wordmark, KPI values
Inter	UI / Body	Nav labels, form labels, body copy, button text
JetBrains Mono	Data / Precision	Timecodes, bitrate values, dB readings, FPS, resolution strings, version numbers

Type Scale

9px	– Micro labels (badge text, version, signal indicators)	– Inter or
10px	– Small labels (section headers, instrumentation strip)	– Inter 600
11px	– Secondary body (status text, timestamps)	– Inter 400
12px	– Primary body (form labels, nav items, settings values)	– Inter 400
13px	– Nav labels, panel headings	– Inter 500
15px	– Page titles (old style – now replaced by 12px uppercase)	– Space (
20px	– KPI values	– Space Gr
28px+	– Hero numbers (peak viewers, revenue)	– Space Gr

Section Header Pattern

All panel section headers follow this exact pattern:

```

<span style={{
  fontSize: 10,
  fontWeight: 600,
  color: "rgba(255,255,255,0.4)",
  letterSpacing: "0.1em",
  fontFamily: "'Inter', sans-serif",
  textTransform: "uppercase"
}}>
  SECTION NAME
</span>

```

Page Title Bar Pattern (Consistent Across All Screens)

Every screen has a top bar with this exact structure:

```

<div style={{ minHeight: 46, background: "#0D0E12", borderBottom: "1px solid #0D0E12" }}
  {/* Page name – all caps, Space Grotesk Bold */}
  <span style={{ fontFamily: "'Space Grotesk', sans-serif", fontWeight: 700 }}>
    PAGE NAME
  </span>

  {/* Vertical divider */}
  <div style={{ width: 1, height: 16, background: "rgba(255,255,255,0.1)" }}>
  </div>

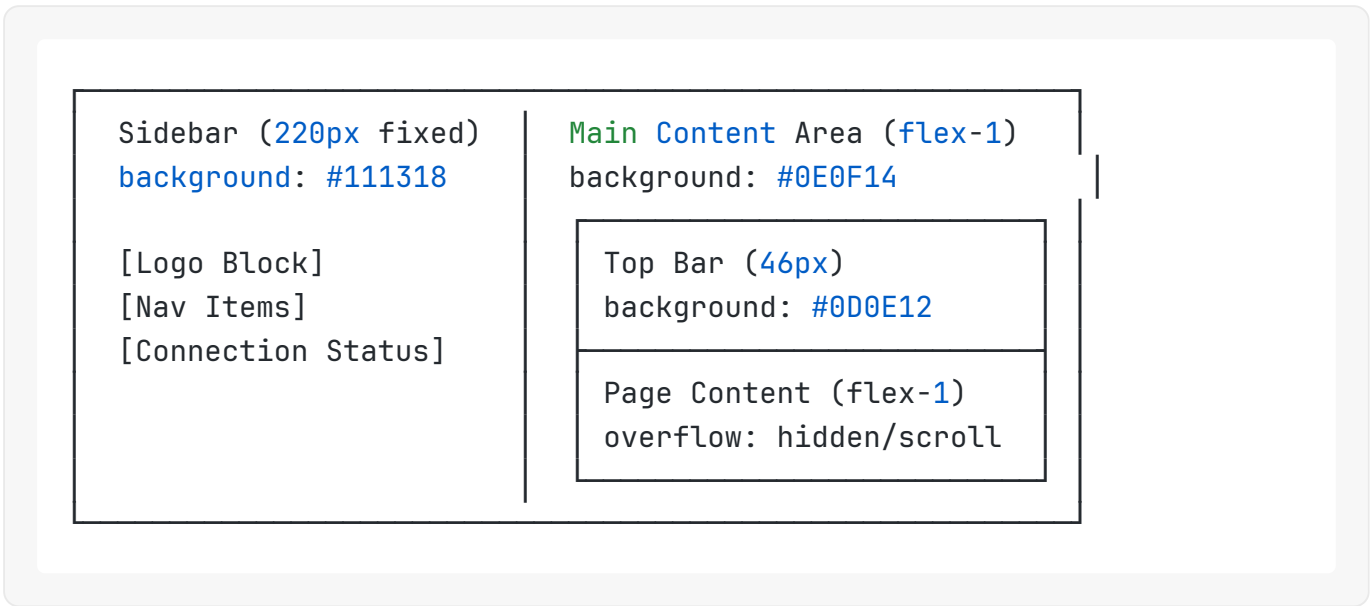
  {/* Subtitle or breadcrumb – JetBrains Mono, dim */}
  <span style={{ fontSize: 10, color: "rgba(255,255,255,0.3)", fontFamily: "JetBrains Mono" }}>
    subtitle / context
  </span>

  {/* Right side: status indicators + actions */}
</div>

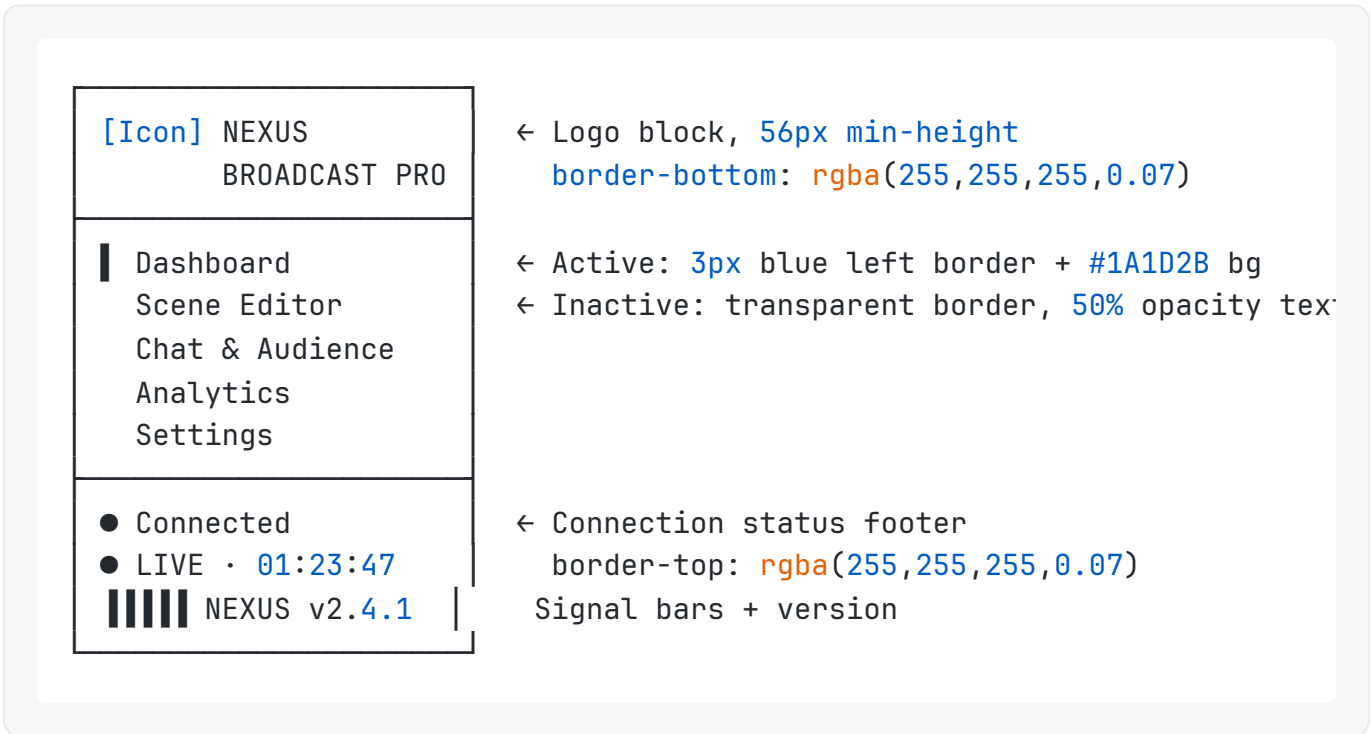
```

4. Layout Structure

Global Shell



Sidebar Anatomy (220px)



Sidebar nav item active state:

```
style={{ background: "#1A1D2B", borderLeft: "3px solid #3B82F6", paddingL
// Icon: strokeWidth 2 (active) vs 1.5 (inactive)
// Text: fontWeight 500 (active) vs 400 (inactive), color white vs white/!
```

5. Screen-by-Screen Specification

Screen 1: Dashboard (/)

Layout: 3-column: [Center: Preview + Scenes + Mixer] | [Right: Stream Status 300px]

Top Bar extras: LIVE badge + timecode + resolution + fps + codec + SIG OK indicator

Program Output Panel:

- Aspect ratio: $\frac{16}{9}$, background #0A0B0F
- SVG grid overlay: `rgba(255,255,255,0.06)` opacity, 40px grid
- 4 corner bracket markers: `rgba(59,130,246,0.6)` blue, 16×16px L-shapes
- Center placeholder: Monitor icon + resolution string in JetBrains Mono
- **Instrumentation strip** below canvas: timecode (cyan) | scene name | SRC ACTIVE (green) | viewer count (blue) | bitrate (green/amber)

Scene Switcher:

- Horizontal scrollable row of 110×70px thumbnails
- Active scene: 2px #3B82F6 border + `box-shadow: 0 0 12px rgba(59,130,246,0.3)` + red LIVE badge
- Inactive: `rgba(255,255,255,0.08)` border

Audio Mixer:

- 4 channels: Mic, Desktop, Music, Alert
- Each channel: VU meter (20 bars, green → amber → red threshold), dB readout in JetBrains Mono, S/M buttons
- Master volume slider with dB display

Stream Status Panel (right, 300px):

- LIVE on YouTube indicator
 - Bitrate: large value in blue + sparkline SVG chart
 - Viewers + Uptime in 2-column grid
 - Stream Health: CPU/GPU/Network progress bars with color-coded values
 - END STREAM button: full-width, #EF4444 background, bold white text
-

Screen 2: Settings (/settings)

Layout: [Left tab nav 180px] | [Content area flex-1] | [Bottom action bar]

Tab nav: background: #111318 , active tab: 3px blue left border + #1A1D2B bg

Tabs: General, Stream, Output, Video, Audio, Hotkeys, Advanced, Plugins

Stream Tab content:

- Platform selector: pill buttons (Twitch/YouTube/Facebook/Custom RTMP), active uses platform color
- Stream Key: password input + Show/Hide + “Get Key ↗” button in blue
- Server: dropdown select
- Encoder: NVENC / QSV / x264 toggle buttons, active in blue
- Encoder params: Rate Control, Bitrate Target/Min, Keyframe Interval — 4-column grid
- Output: Recording path + Browse, Format buttons (MKV/MP4/MOV), Replay Buffer toggle
- Video: Canvas Resolution, Output Resolution dropdowns + FPS toggle (30/60/120)

Form row pattern:

```
// Label (160px fixed width) + control (flex-1)
// Divider: borderBottom: "1px solid rgba(255,255,255,0.05)"
```

Bottom action bar: Cancel (ghost) + Save Settings (blue primary) — fixed to bottom

Screen 3: Chat & Audience (/chat)

Layout: [Left panel 220px] | [Center chat flex-1] | [Right panel 280px]

Left panel:

- Mini program preview (¹⁶/₉ with timecode overlay)
- Quick Actions grid: Clip, Raid, Host, Poll (2×2) + Marker (full width)

Chat center:

- Header: platform color dots (T/Y/F) + viewer count badge
- Message types:
 - Regular chat: transparent background
 - Donation: `rgba(245,158,11,0.1)` bg + amber border
 - Sub: `rgba(145,70,255,0.1)` bg + purple border
- Message anatomy: `[timestamp] [MOD/SUB badge] [username:] [message]`
- MOD badge: green bg/border; SUB badge: purple bg/border
- Input: dark input + emoji button + send button (blue)
- Mod tools row: shield, ban, timeout, slow mode, clear icons

Right panel:

- Activity Feed: follow/sub/donation/raid events with colored icons
- Polls & Predictions: progress bars with percentage + countdown timer
- Chat Commands: grid of `!command` buttons with descriptions

Screen 4: Analytics (`/analytics`)

Layout: Full-width scrollable content, 3-row structure

KPI Cards (4-column grid):

- Each card: colored top-left icon + metric label + large value (Space Grotesk 700) + delta badge
- Delta: green for positive (`+18.7% vs yesterday`), red for negative
- Card border uses accent color at 30% opacity: `border: 1px solid ${color}30`

Viewer Chart:

- Full-width recharts AreaChart/LineChart
- LIVE DATA badge (red pulse) in header
- Instrumentation strip: PEAK / AVG / CURRENT values in JetBrains Mono

- Time range selector: 5m / 15m / 1h / 3h / 6h / 12h pill buttons

Middle row (2-column):

- Stream Health: list of metrics (Bitrate, CPU, GPU, FPS, Dropped Frames) with colored right-aligned values
- Audience Breakdown: donut chart + platform legend + Top Countries bar chart

Recent Sessions table:

- Columns: Date, Duration, Peak Viewers (blue), Avg Viewers, Revenue (green), Quality Score (badge)
- Quality Score badges: Excellent (green), Good (blue), Fair (amber)

Screen 5: Scene Editor (/scenes)

Layout: [Canvas + toolbar flex-1] | [Right inspector 300px]

Top bar extras: Project name chip (blue border) + live telemetry (LIVE/REC timers, CPU%, FPS, KB/s)

Canvas workspace:

- Background: #0A0B0F
- SVG grid: 60px grid, rgba(255,255,255,0.04) opacity
- Crosshair at center: 20px lines, rgba(59,130,246,0.25)
- Selected source: blue dashed border + 8 resize handles (corner + edge)
- Canvas metadata strip (bottom): resolution · aspect ratio · fps | scene name · source count | SAFE AREA: ON
- Corner bracket markers: rgba(59,130,246,0.6) L-shapes

Right toolbar (vertical icon strip): Move, Scale, Crop, Rotate, Align, Lock — icon buttons with hover states

Sources panel (right, 300px):

- Header: SOURCES label + Add (+) + More (⋮) buttons
- Source rows: drag handle + visibility eye + lock + type icon + name + more menu
- Active source: #1A1D2B background + blue left accent
- Source types: Game Capture, Webcam, Text Overlay, Logo Image, Alert Overlay

Source Properties panel:

- Header: “SOURCE PROPERTIES: [NAME]” in uppercase
- Position (X/Y), Size (W/H), Rotation, Opacity — numeric inputs in JetBrains Mono
- Chroma Key toggle (green when active)
- Color picker + Similarity slider
- Flip Horizontal / Flip Vertical buttons
- Crop inputs (Top/Bottom/Left/Right) + Reset Crop button

Transitions rail (bottom):

- Transition types: Cut (active, blue), Fade, Slide, Wipe, Stinger
- Duration input (ms) + Apply Transition button

6. Reusable Component Patterns

Panel / Card

```
<div style={{
  background: "#111318",
  border: "1px solid rgba(255,255,255,0.08)",
  borderRadius: 8 // 0.5rem
}}>
  {/* Panel header */}
  <div style={{ borderBottom: "1px solid rgba(255,255,255,0.06)", padding
    <span style={{ fontSize: 10, fontWeight: 600, color: "rgba(255,255,255,0.8)"
      PANEL TITLE
    </span>
  </div>
  {/* Panel content */}
</div>
```

LIVE Badge

```
<div style={{ display: "flex", alignItems: "center", gap: 6, background: '
  <div style={{ width: 6, height: 6, borderRadius: "50%", background: "#E
  <span style={{ fontSize: 10, fontWeight: 700, color: "#EF4444", letterSp
</div>
```

Status Indicator (inline)

```
// Green = good, Red = bad, Amber = warning
<div style={{ display: "flex", alignItems: "center", gap: 6 }}>
  <div style={{ width: 6, height: 6, borderRadius: "50%", background: "#2
  <span style={{ fontSize: 10, color: "rgba(255,255,255,0.4)", fontFamily
    SIG OK
  </span>
</div>
```

Instrumentation Strip (below canvas/chart)

```
<div style={{ display: "flex", justifyContent: "space-between", padding:
  <div style={{ display: "flex", gap: 12 }}>
    <span style={{ color: "rgba(255,255,255,0.25)" }}>TC</span>
    <span style={{ color: "#06B6D4" }}>01:23:48:01</span>
    <span style={{ color: "rgba(255,255,255,0.1)" }}>|</span>
    <span style={{ color: "#22C55E" }}>SRC ACTIVE</span>
  </div>
  <div style={{ display: "flex", gap: 12 }}>
    <span style={{ color: "rgba(255,255,255,0.25)" }}>BITRATE</span>
    <span style={{ color: "#22C55E", fontWeight: 600 }}>8500 kbps</span>
  </div>
</div>
```

Primary Button

```
<button style={{
  background: "#3B82F6",
  color: "#fff",
  border: "none",
  borderRadius: 6,
  padding: "8px 16px",
  fontSize: 12,
  fontWeight: 600,
  fontFamily: "'Inter', sans-serif",
  cursor: "pointer",
  transition: "transform 0.1s ease, opacity 0.1s ease",
  // On :active: transform: scale(0.97)
}}>
  Save Settings
</button>
```

Danger Button (End Stream)

```
<button style={{
  background: "#EF4444",
  color: "#fff",
  width: "100%",
  padding: "10px",
  borderRadius: 6,
  fontSize: 13,
  fontWeight: 700,
  letterSpacing: "0.04em",
  border: "none",
  cursor: "pointer",
}}>
  ■ END STREAM
</button>
```

Toggle Switch

```
// Width: 36px, Height: 20px
// On: background #3B82F6, thumb left: 18px
// Off: background rgba(255,255,255,0.15), thumb left: 2px
// Thumb: 16×16px white circle, transition: left 0.2s ease
```

Text Input

```
<input style={{
  background: "#0E0F14",
  border: "1px solid rgba(255,255,255,0.12)",
  borderRadius: 6,
  color: "#fff",
  fontSize: 12,
  fontFamily: "'Inter', sans-serif",
  padding: "6px 12px",
  outline: "none",
  // On focus: border-color: rgba(59,130,246,0.5)
}} />
```

VU Meter (Audio)

```
// 20 bars, each 3px wide, 1px gap
// Heights: 4px + (i % 3)px – slight variation for organic feel
// Color thresholds: bars 0-13 = #22C55E, 14-16 = #F59E0B, 17-19 = #EF4444
// Inactive bar: rgba(255,255,255,0.1)
// Update at 60fps via requestAnimationFrame or setInterval(16ms)
```

Health/Progress Bar

```
// Container: height 4px, background rgba(255,255,255,0.08), border-radius: 2px;
// Fill: width = value%, color = green/amber/red based on threshold, transition: width 0.1s ease-out;
```

7. Animation & Motion

```
/* LIVE pulse – used on all red dot indicators */
@keyframes pulse {
  0%, 100% { opacity: 1; transform: scale(1); }
  50% { opacity: 0.5; transform: scale(1.3); }
}

/* Standard easing for all UI transitions */
--ease-out: cubic-bezier(0.23, 1, 0.32, 1);

/* Timing guidelines */
button hover/active: 100–160ms ease-out
nav item hover: 120ms ease-out
panel entrance: 150ms opacity + translateY(4px)
modal/drawer: 200–300ms ease-out
chart draw-in: 600ms ease-out
bitrate/health bars: 500ms ease transition on width
VU meters: 50ms (near-instant, feels live)
```

Button press feedback (required on all interactive buttons):

```
button:active { transform: scale(0.97); transition: transform 0.1s ease-out; }
```

8. Canvas / Workspace Instrumentation Rules

Any large empty workspace (program output preview, scene editor canvas, chart area) **must** include:

1. **SVG grid overlay** — subtle `rgba(255,255,255,0.04-0.06)` grid pattern
 2. **Corner bracket markers** — 4 L-shaped brackets in `rgba(59,130,246,0.6)` at corners
 3. **Metadata strip** — bottom bar with resolution, scene name, status in JetBrains Mono
 4. **Center placeholder** — icon + label when no content is loaded
 5. **Crosshair** (scene editor only) — thin cross at canvas center in `rgba(59,130,246,0.25)`
-

9. Spacing & Radius

```
Panel padding:      12px (p-3)
Section gap:        12px (gap-3)
Inner element gap:  8px (gap-2)
Border radius:      8px (rounded-lg) for panels
                   6px (rounded-md) for buttons/inputs
                   4px (rounded) for badges/chips
                   9999px for toggles/pills

Top bar height:     46px minimum
Sidebar width:       220px (fixed, never collapses on desktop)
Right panel width:   280-300px (fixed)
```

10. Do's and Don'ts

Do

- Use `#0E0F14` as the root background — never pure `#000000`
- Use `JetBrains Mono` for all numeric readouts, timecodes, and technical values
- Keep the LIVE badge visible on every screen while streaming
- Add instrumentation strips to any large empty area
- Use `rgba()` for borders — never opaque colored borders on dark surfaces
- Keep Nexus Blue `#3B82F6` as the **only** primary interaction color
- Use `Space Grotesk` for page titles and KPI numbers
- Make all top bars exactly 46px tall with `background: #0D0E12`

Don't

- Don't use purple gradients, rounded hero sections, or centered landing-page layouts
- Don't use `Inter` for timecodes or numeric data — always `JetBrains Mono`
- Don't make panels float freely — they should feel anchored to a grid
- Don't use platform colors (Twitch purple, YouTube red) as dominant panel colors
- Don't add decorative illustrations or icons that don't serve a functional purpose
- Don't use `border-radius > 8px` on panels — keep corners sharp and professional
- Don't animate layout properties (`width` , `height` , `margin`) — only `transform` and `opacity`
- Don't use white text on anything brighter than `#1A1D2B` — contrast will fail